Law of Sines and Law of Cosines

ANSWER KEY

Section 1: The Law of Sines

1 State the Law of Sines for any triangle ABC.

a. $\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$ ✓

b. $a^2 = b^2 + c^2 - 2bc\cos A$

c. $a\sin A = b\sin B = c\sin C$

d. $\frac{a}{\cos A} = \frac{b}{\cos B}$

The Law of Sines states that the ratio of each side length to the sine of its opposite angle is equal for all three vertices.

2 In $\triangle ABC$, $A = 30^\circ$, $B = 45^\circ$, and $a = 8$. Find the exact length of side b.

a. $8\sqrt{2}$ ✓

b. $4\sqrt{2}$

c. $8\sqrt{3}$

d. 16

$\frac{8}{\sin 30^\circ} = \frac{b}{\sin 45^\circ} \Rightarrow \frac{8}{1/2} = \frac{b}{\sqrt{2}/2} \Rightarrow 16 = \frac{2b}{\sqrt{2}} \Rightarrow b = 8\sqrt{2}$.

3 Under which given information cases can the Law of Sines be directly applied to begin solving a triangle?

a. **AAS or ASA** ✓

b. SAS or SSS

c. SSS only

d. Right triangles only

The Law of Sines requires at least one known angle-side opposite pair, which is always provided in AAS and ASA cases.

Section 2: The Ambiguous Case (SSA)

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4 In the ambiguous SSA case with acute angle A, what does $h = b\sin A$ represent?

a. **The altitude from vertex C to side c ✓**

b. The perimeter of the triangle

c. The hypotenuse

d. The length of side a

$h = b\sin A$ is the perpendicular distance (height) from vertex C to the opposite line.

5 If $A = 30^\circ$, $b = 10$, and $a = 4$, how many triangles can be formed?

a. **0 (no triangle exists) ✓**

b. 1 triangle

c. 2 triangles

d. Infinitely many

$h = 10\sin(30^\circ) = 5$. Since $a = 4 < h = 5$, the side is too short to reach the opposite baseline, so no triangle can be formed.

6 If $A = 30^\circ$, $b = 10$, and $a = 7$, how many triangles can be formed?

a. **2 triangles ✓**

b. 1 triangle

c. 0 triangles

d. Cannot be determined

$h = 10(0.5) = 5$. Since $h < a < b$ $(5 < 7 < 10)$, side a can swing to two different locations, forming exactly 2 triangles.

Section 3: The Law of Cosines

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7 State the Law of Cosines to solve for side c given sides a, b and included angle C.

a. $c^2 = a^2 + b^2 - 2ab\cos C$ ✓

b. $c^2 = a^2 + b^2 + 2ab\cos C$

c. $c = a\cos B + b\cos A$

d. $c^2 = a^2 + b^2 - ab\sin C$

The Law of Cosines is the generalized Pythagorean theorem: $c^2 = a^2 + b^2 - 2ab\cos C$.

8 In $\triangle ABC$, $a = 5, b = 7$, and $C = 60^\circ$. Find the exact length of side c.

a. $\sqrt{39}$ ✓

b. $\sqrt{74}$

c. $\sqrt{49} = 7$

d. 6

$c^2 = 5^2 + 7^2 - 2(5)(7)\cos(60^\circ) = 25 + 49 - 70(0.5) = 74 - 35 = 39 \Rightarrow c = \sqrt{39}$.

9 In $\triangle ABC$, $a = 3, b = 5, c = 7$. Find the exact value of $\cos C$.

a. $-1/2$ ✓

b. $1/2$

c. $-1/5$

d. $-3/7$

$\cos C = \frac{a^2 + b^2 - c^2}{2ab} = \frac{9 + 25 - 49}{2(3)(5)} = \frac{34 - 49}{30} = \frac{-15}{30} = -\frac{1}{2}$.

Section 4: Area of an Oblique Triangle

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10 State the sine formula for the area of $\triangle ABC$ given sides a and b and angle C.

a. $\text{Area} = \frac{1}{2}ab\sin C$ ✓

b. $\text{Area} = ab\sin C$

c. $\text{Area} = \frac{1}{2}ab\cos C$

d. $\text{Area} = \frac{a+b}{2}\sin C$

Using altitude $h = b\sin C$, base a gives $\text{Area} = \frac{1}{2}ab\sin C$.

11 Find the exact area of $\triangle ABC$ if $a = 6$, $b = 10$, and $C = 30^\circ$.

a. 15 ✓

b. 30

c. $15\sqrt{3}$

d. 7.5

$\text{Area} = \frac{1}{2}(6)(10)\sin(30^\circ) = 30(1/2) = 15$.

12 Heron's formula states $\text{Area} = \sqrt{s(s-a)(s-b)(s-c)}$. Find the area of a triangle with sides 5, 6, and 7.

a. $6\sqrt{6}$ ✓

b. $12\sqrt{3}$

c. 18

d. 14

Semi-perimeter $s = \frac{5+6+7}{2} = 9$. Area
$= \sqrt{9(9-5)(9-6)(9-7)} = \sqrt{9(4)(3)(2)} = \sqrt{216} = \sqrt{36 \times 6} = 6\sqrt{6}$.

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